Analysis of the Effects of Housing Permits on Population Growth from 2010-2022: Propensity Score Matching

Introduction

Population growth is one of the most important factors reflecting how communities change over time, as it can affect (and be affected by) labor market trends, infrastructure quality, public services, and demand for housing, making it crucial to understand the factors that have the strongest effects on population growth. One of the most well studied variables is the availability of housing, as the availability of housing can not only affect the price of housing (thereby affecting the cost of living) but also the physical capacity for a city to hold more people and hire more people within the area itself. House permits, which are a very strong correlate of housing construction, give a clear indication of changes in housing supply, and can be used to examine the relationship between construction and population growth. Additionally, house permitting is affected by a multitude of factors, ranging from the amount of land left for zoning, to the regulatory environment in the area. However, population growth is affected by many socioeconomic, demographic, and labor market factors, making the relationship between permitting and population growth far too complicated for simple statistical modeling.

The goal of this project is to evaluate the relationship between house permitting (a strong correlate of house construction) and population growth across U.S. counties from 2010-2022. Using county-level housing, socioeconomic, demographic, and labor market data, this project applies multiple statistical approaches, and machine learning methods to determine the relationship from different perspectives, including linear relationships, nonlinear prediction methods, growth forecasting, and causal inference. This specific report uses causal inference

testing to determine whether house permitting has a causal effect on population growth. Propensity score matching is used to address the limitations of observational data by creating groups of counties which have similar values for all variables except house permits and population growth.

Data

In order to determine the effect of housing permits on population growth, there were many covariates that were used for different reasons. These included the total permit rate per 1,000 people (obtained from the total number of permits in the county, divided by the population, multiplied by 1,000), as well as the number of single family permits per 1,000, the number of multifamily permits per 1,000 (specifically 2-4 unit permits), and the number of permits for buildings with at least 5 units (per 1,000). Each permit-rate variable was used as the main independent variable for its own regression, while the total permit rate was used for every other type of model. Additionally, the remaining housing variables included county-level homeownership rates, the growth rate in occupied housing units (which was lagged by a year), rent burden (the percentage of people paying more than 30% of their income in rent), and median house value. The labor market and economic characteristics that were used included median household income and the employment population ratio. The demographic variables that were used were the percent of people in a county who are white, the percent of people in a county who are black, the percent of people born outside the US, the change in the percent of people born outside the US (essentially functioning as net migration), as well as the county's median age. Lastly, I also used population density for the counties, which functions as a ceiling on how many

people can live in an area before geographic constraints inhibit housing construction (albeit, for many counties, that limit hasn't been reached).

Data Preparation

With the exception of the permit rate, I gathered all of the data from the American Community Survey's 5 year estimates. I used 5 year estimates instead of 1 year estimates because ACS data becomes very noisy when applied to small counties due to their small sample sizes for data collection. For the permit rate, I gathered information on total permits from the Department of Housing and Urban Development, which has the data from 1980-2022, after which I divided the number of each permit type by the population, and I multiplied them by 1,000 to get the per 1,000 rate. I used data from 2010-2022 for almost all of the variables because much of the ACS data starts in 2010, while the housing permit data set runs up to 2022. I also used 2023 data for population, foreign-born population, and the number of occupied housing units. These were used to create growth rates which were mostly used for lags so that I could use the 2022 data to determine trends. The data for the variables (from ACS data) was collected using API keys, and they were merged using SQL joins (mostly inner joins), with some variable transformations and calculations being performed in SQL using window functions, with other transformations being performed either in Excel or in Python workbooks.

For all of the models, I had to make some decisions about what data to include and what to remove. Firstly, since the COVID-19 pandemic falls within the time frame of this project, there are some years where population growth is significantly lower as a result of a crisis that can't easily be seen in the data. The period when this issue was most apparent was 2020-2021, as population growth was significantly lowered from 2020 to 2021 as hundreds of thousands died

from the virus. While even more people died in 2021, this ended up having less of a deleterious effect on population growth because of the substantial increases in immigration during 2021. Since corresponding population growth was the variable of choice, population growth from 2020 to 2021 was associated with the values for each of the variables in 2020. As a result of the issues that 2020 would present for the data, I made the decision to remove the 2020 data from consideration, as well as its corresponding population growth. Although this made models like forecasting far more difficult, the alternative was worse as a whole year's worth of data would be of incredibly poor quality.

Additionally, since there were some very noticeable outliers which could heavily alter the relationship between an independent and dependent variable, I tended towards removing them. This was firstly done by removing counties with a population below 2,500 as population growth figures are incredibly noisy at such low levels, since even a couple families deciding to move out can have a substantial effect on population growth. Additional outliers which were removed included counties which allowed more than 20 permits per 1,000 (a sample of roughly 30 observations), counties which had population growth rates above 8%, and below -5%, (a sample of roughly 40 observations), and counties which saw their number of available housing units either increase by more than 15% in one year, or decrease by more than 12.5% in one year (a sample of roughly 160 observations). While these removed a meaningful amount of data (roughly 0.6% of the data in the sample), even small amounts of outliers were capable of changing the best relationship between an IV and a DV from a linear relationship to a logarithmic one, which is especially problematic for something like regression modeling, where the wrong relationship being specified can make models perform far worse.

Variables

Besides the house permit rate, which is the main independent variable for the project, the housing variables in question were used to communicate how affordable housing is in some way, or to communicate the availability of housing in full. Median house value, rent burden, and homeownership rate are used for different reasons to tell how affordable housing is in terms of the price of housing, the household level constraints from owning housing, or how many people own homes (because many people who don't own houses would own them if they could). The growth in housing units is a reflection of the housing stock at any given time, and even though it could be highly correlated with permitting frequency, it isn't, even when it's lagged.

The employment population ratio was used because it's a simple reflection of how many people work and is superior to the employment rate in some respects because it can better account for an area having lots of discouraged workers, retirees, or limited employment opportunities. Median household income was used to determine how wealthy the average person is in an area.

When it came to demographic variables, I used the three groups (white people, black people, and foreign born people) for different reasons. There is a well documented history of white people moving out of urban areas and into suburban areas when they become more racially diverse, especially with respect to black people moving into an area, and this pattern has been present as suburbs have become more diverse as well (even in middle class neighborhoods), so a higher share of black people in an area may mean that white people move to a nearby county[1][2][3]. I used the white percentage to convey the level of racial homogeneity which may be a correlate of increased housing values for the same reason that an increased presence of minorities may be a correlate of decreased house values. I used the foreign born share not only as

a variable related to diversity, but also as a way to show a crude form of net migration which is itself a massive component of population growth that may not be captured by the data in any other way, especially relevant because the reason for immigrants coming to a certain county aren't likely to be properly predicted from the data. While immigrant flows are somewhat correlated with local labor market conditions, the nature of a labor market can only partially be inferred from the data in question[4].

Lastly, I used population density to communicate the extent to which a county has already been urbanized, its ability to get more permits approved, and its capacity to become more densely populated. I chose this as a measure instead of an urban vs rural percentage because the definition of an urban area has become more restrictive with time, including through an increase in the minimum population requirement.

Methodology

In order to make the proper models, it's critical to ensure that each variable is correctly specified. This is done by checking for the shape of each variable when plotted against population growth, along with the R^2 and the RMSE (root mean squared error) having a minor role in determining whether a transformation is best. However, if the relationship between a given variable and population growth looks linear, I defaulted to using a linear transformation to avoid the risk of fixating too much on small deviations in R^2 and RMSE. Since population growth is more or less normally distributed, it does not require a logarithmic transformation. Therefore, the other variables are measured against a dependent variable which has a linear specification. Ultimately, for most variables, a linear specification was adequate. This includes permits per 1,000 (which applies to all 4 of the permit variables), homeownership rate, the

employment population ratio, the change in occupied housing units, the change in the percent of foreign born people in a county, the rent burden, the median house value, and the median age in years, however, the two interaction terms cannot be used for the causal inference model because they make the model perform far worse. Logarithmic specifications were used for the foreign born population share, as well as population density. A quadratic specification was used for the percent of people in a county who are white, and a cubic specification was used for the percent of people in a county who are black, while a reciprocal specification was used for the median household income.

The causal inference model chosen for this project was propensity score matching. Propensity score matching is a statistical method applied to observational data in order to mitigate selection bias by taking individuals or observations from different groups and matching them together[5]. The purpose is to take observational data and use it to be able to mitigate the conditions of a randomized control trial as best as you can. Observational data is flawed because being assigned the treatment is not random, people may be more likely to use a certain drug, or consume a certain food based on factors like their age, health, income, etc. Likewise, individual counties may be more likely to permit more housing construction based on factors related to how often people are moving there, and the labor market conditions (assuming the right environment exists to be able to permit houses to be built, which was not easily able to be measured in the data).

Propensity scores refer to the probability that an individual receives a treatment conditional on all other covariates. Propensity scores are generated from logistic regressions that are used to estimate the probability of receiving a treatment. In randomized trials, treatments would be randomly assigned, but in observational data, those covariates can help you understand

which observations would be more likely to get a certain treatment. In the case of this project, that treatment comes from receiving more permits than the average county. After calculating propensity scores, the next steps are to match observations with similar propensity scores, then to check for covariate balance where you see whether two groups have relatively similar values for all of the covariates, at which point, you can determine the average treatment effect. The form of propensity score matching I used uses K-Nearest Neighbors where it matches each treated observation with an untreated observation that has a similar propensity score, meaning that it looks for two observations that are equally likely to be treated with only one of them actually being treated, and it subsequently compares the values of the dependent variable for both of them, with the only difference being in terms of the treatment "assignment." Matching quality is judged using standardized mean differences, which compare the mean values of variables between two groups. Values below 0.1 indicate strong covariate balance, meaning the difference in outcomes can be attributed to differences from receiving the treatment or not.

The downsides of the model come from the fact that the causal interpretation is only as good as the variables in the model itself. If there's omitted variable bias, the model can't fix that because even performing perfect balancing won't fix the fact that the model isn't accounting for all relevant factors. Additionally, since the difference between getting the "treatment" (being a high permit county) is not random, counties having low permit rates is somewhat predictable based on other information from variables, therefore many of those counties won't be matched to a similar county that did get a treatment. As a result, there is a significantly lower sample size when applied to causal inference methods. However, since the sample size of the data is extremely high and approaching 40,000 observations, that is far less of a problem in this context.

The models were graded using ROC-AUC, which measures the ability of a model to distinguish between true positives, and false positives, with a value closer to 1 being better[6]. For the causal inference model, I made two models, one for the linear versions of every variable, and one using all of the requisite transformations to fit the data the best.

Results

For the causal inference model based on linear transformations of the data, the average treatment effect was 0.5036, meaning that a county going from below-median to above-median permitting leads to a 0.5036% increase in population growth. All of the covariates like household income, homeownership rate, employment/population ratio, etc had standardized mean differences below 0.1, except for foreign born percent, which had a standardized mean difference of 0.11. For the causal inference model which allowed for the requisite nonlinear transformations of the data, the average treatment effect was 0.5132, while the model performed slightly better in terms of ROC-AUC. Unlike the model based on linear transformations, the standardized mean differences were all below 0.1.

Discussion

Based on all of the tests for model strength, the two models performed quite well, as both had standardized mean differences below 0.1 for almost every variable, meaning that the observations in the control group and experimental group were very similar in every facet besides permitting rate. Additionally, the ROC-AUC values were both high, as the linear model had a ROC-AUC value of 0.8073, and the transformed model had a value of 0.8090. Both models said that going from permitting less housing construction than the median county to

permitting more construction leads to the county's population growth increasing by roughly 0.5% yearly. However, even though rigorous methods are used to make the model as good at estimating causality as is possible, the model is still only as good as what is given to it. If there are omitted variables, it won't be able to fix that just because it's good at matching similar counties, and since it's infeasible to include every variable in existence, it's important to know whether the missing variables would be able to undermine the relationship between permit rate and population growth. In the case of this model, the main missing variables are tax burden, public service quality, climate conditions, homeowner insurance, worker protections, transportation quality, infrastructure quality, school quality, and job availability (essentially the diversity of jobs that are readily available in a city).

In practice, higher taxes, weaker public services, poorer climate, less space for building houses, more costly homeowner insurance, inadequate worker protections, poor transportation systems and public transit availability, poor infrastructure, poor school quality, and a job market that isn't diverse are likely to push people away from a county. A place which allows for more houses to be built is likely to have a more laissez-faire local economy, which may mean lower taxes, worse public services, more space for building housing, less costly homeowner insurance, and fewer policies to protect workers. The remaining factors cannot easily be said to be systematically affected by (or correlated with) house permitting. While this is speculative, it's important to recognize that these variables do not clearly point to the propensity matching results overestimating or underestimating the actual effect of permitting.

Additionally, the model R^2 was at around 0.45 for even the worst performing models throughout the entirety of the project, in spite of the models measuring population growth (including in very small counties), which is a difficult variable to model because it's not

necessarily an indicator of quality of life. Furthermore, as you add more variables, the contributions of each variable to explanatory power are likely to diminish because much of the effect of a new variable has already been captured by prior variables. As an example, when I decided to add a variable for the change in foreign born population as a proxy for net migration, the effect on model performance was almost undetectable in spite of it being an incredibly important factor in determining how much a region will grow or shrink. In fact, adding too many variables may even be detrimental in some cases like regression because of multicollinearity. While a limited improvement in model performance doesn't mean that a variable can't noticeably influence the treatment effect, the omitted variables aren't likely to bias the causal estimate in the same direction to any major degree. Therefore, adding more variables isn't likely to produce a very large change, let alone one that meaningfully affects policymaking perspectives.

However, a major weakness which is present throughout all of this project is reverse causality. Although higher permitting is consistently shown to be strongly related to higher population growth, higher population growth is also very likely to lead to a county approving more permits, and this can't easily be remedied. While an instrumental variable (something which affects population growth only through its effect on permitting) is very helpful, instrumental variables are not only difficult to find, but the logic underpinning their inclusion in a model can be very flawed, and they often do affect dependent variables independently. Additionally, even if a truly robust instrumental variable is present, it may not have enough data to fit the scope of this project, as it would need to consistently be present throughout counties of different sizes for a period spanning roughly 15 years.

Conclusion

Ultimately, while reverse causality is a concern, the main takeaway is that allowing more housing construction contributes to higher population growth, while population growth is likely to incentivize higher permitting rates. As a result of increased population growth, you're likely to see reinvigorated local demand, and increased job growth, potentially increasing the economic influence of many major cities like New York City, Boston, Seattle, and San Francisco. While so-called "superstar cities" have a plethora of reasons explaining why their cost of living may be so high, a major factor is the fact that the cities have relatively low housing stocks. The methods that can be used to facilitate increased housing construction include eliminating minimum lot size requirements that can constrain development opportunities, along with reducing mandatory parking lot capacities in apartment complexes, and striking down laws that only permit single-family housing in an area.

However, these measures may not be sufficient, and more controversial and direct methods to increase construction may be needed, including mandating construction in an area, regardless of local opposition, but with compensation for those whose housing values decrease as a result of increased supply, along with state governments threatening to limit funding allocations to local governments if they fail to permit an acceptable amount of housing to keep prices in check. While the purpose of this project is not to determine which of these policies are most successful or feasible, the findings do show that local permitting has a major effect on population growth, which means that counties struggling to permit enough housing should consider a plethora of policymaking tools in order to reverse trends of population decline as a result of inadequate housing construction.

Appendix

The housing permits dataset was obtained from the department of housing and urban development[7]. The ACS data included the datasets for Selected Social Characteristics, Selected Housing Characteristics, Income in the Past 12 Months, Place of Birth by Poverty Status in the Past 12 Months in the United States, Age and Sex, Hispanic or Latino Origin by Race, Selected Economic Characteristics, and the Annual Geographic Information Table (GEOINFO)[8][9][10][11][12][13][14][15].

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